Chapter 1

Introduction

Very roughly speaking, representation theory studies symmetry in linear spaces. It is a beautiful mathematical subject which has many applications, ranging from number theory and combinatorics to geometry, probability theory, quantum mechanics, and quantum field theory.

Representation theory was born in 1896 in the work of the German mathematician F. G. Frobenius. This work was triggered by a letter to Frobenius by R. Dedekind. In this letter Dedekind made the following observation: take the multiplication table of a finite group G and turn it into a matrix X_G by replacing every entry g of this table by a variable x_g. Then the determinant of X_G factors into a product of irreducible polynomials in $\{x_g\}$, each of which occurs with multiplicity equal to its degree. Dedekind checked this surprising fact in a few special cases but could not prove it in general. So he gave this problem to Frobenius. In order to find a solution of this problem (which we will explain below), Frobenius created the representation theory of finite groups.

The goal of this book is to give a "holistic" introduction to representation theory, presenting it as a unified subject which studies representations of associative algebras and treating the representation theories of groups, Lie algebras, and quivers as special cases. It is designed as a textbook for advanced undergraduate and beginning